
Pricing in the power system

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2 Introduction to pricing

3 Spot market

4 Grid tariffs



About us



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- Industrial Economy and Technology Management
 - Finance, Energy and Environment
- Graduated in 2025

In THEMA:

- Market analysis
- Quantitative modelling and optimization
- Strategy



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In THEMA:

- Market analysis
- Flexibility in power markets
- Quantitative modelling and optimization



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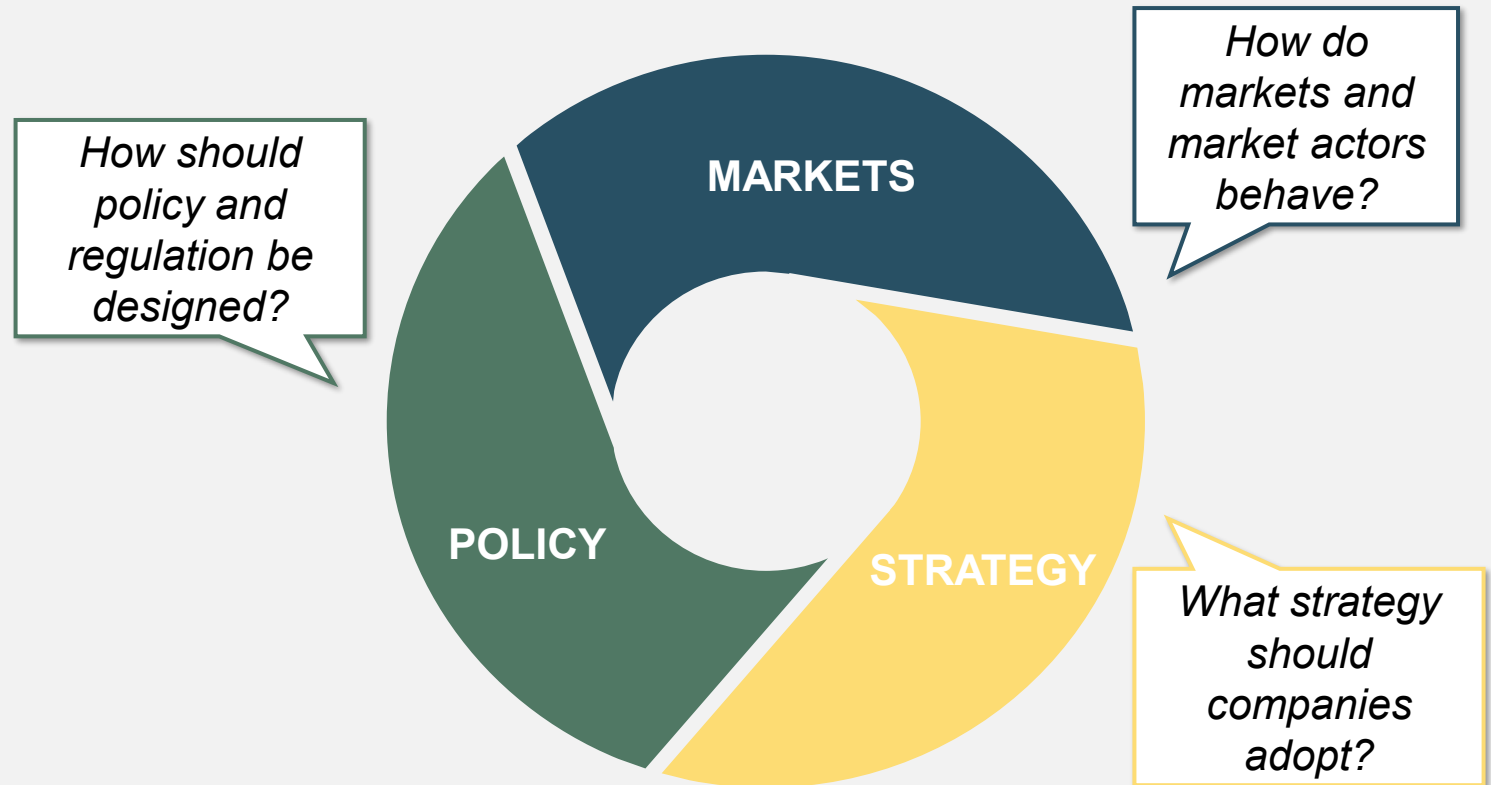
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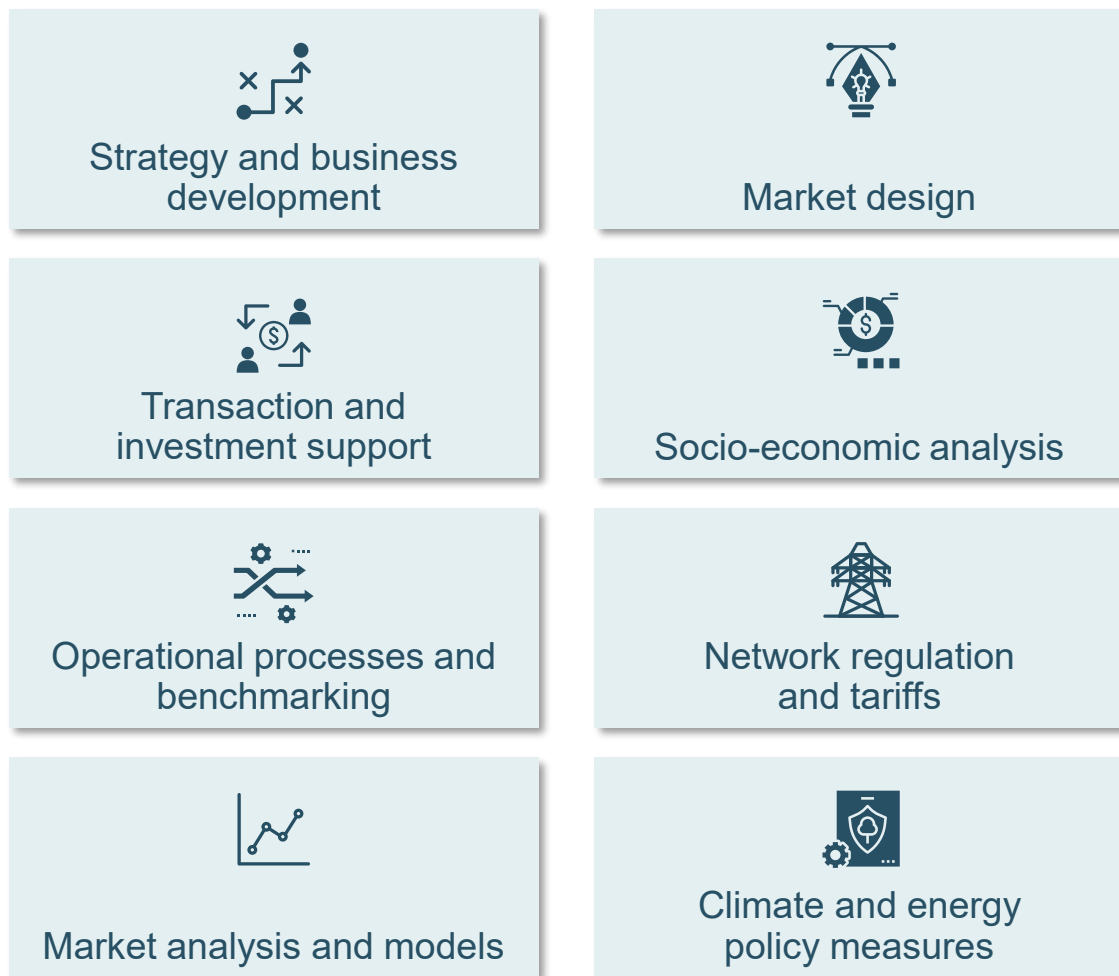
Offices in
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What we do

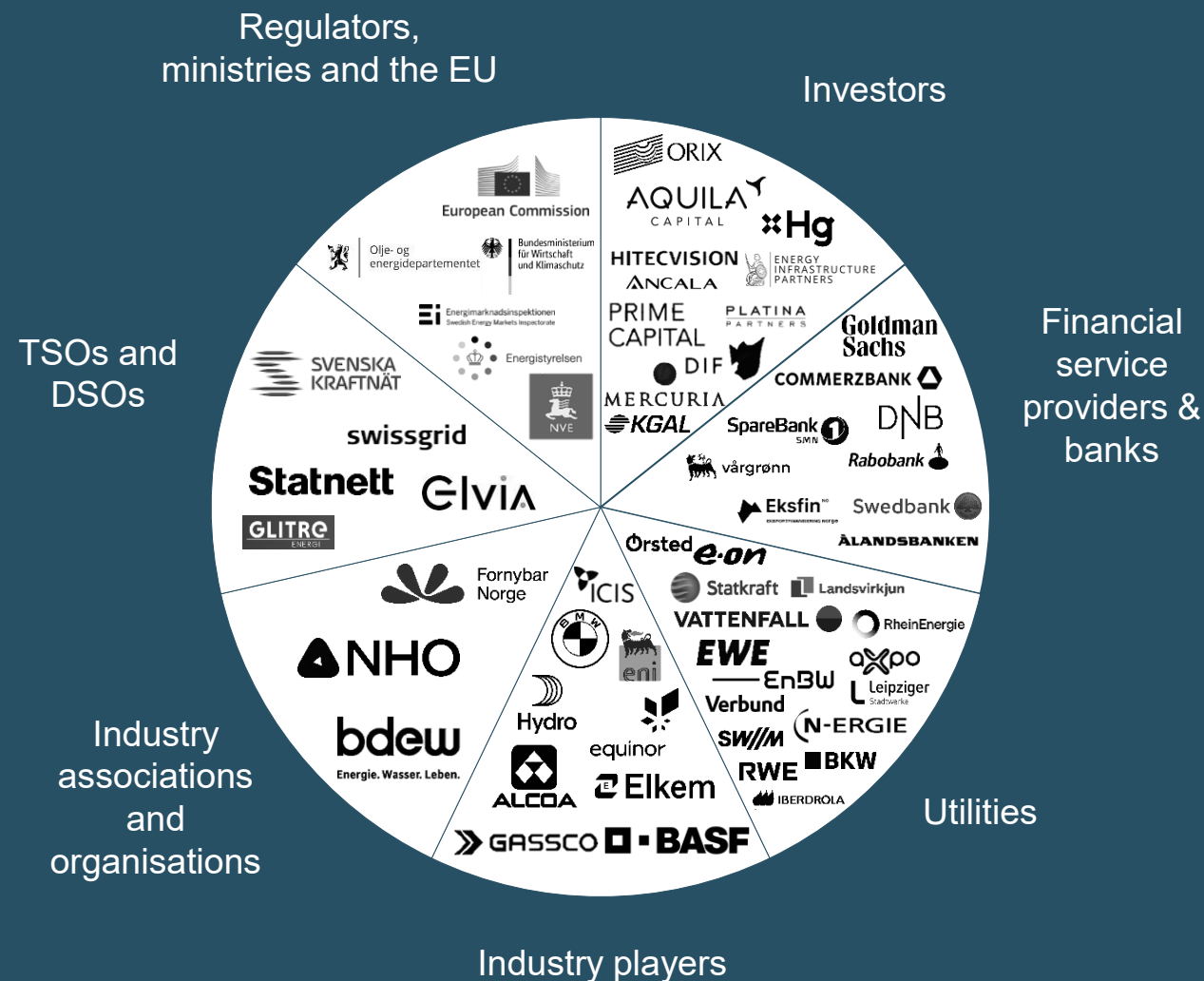
Our work draws on combined expertise in the following aspects of the energy transition



We provide trusted, independent analysis and advice...



...to a diverse range of clients across the value chain



1 About us

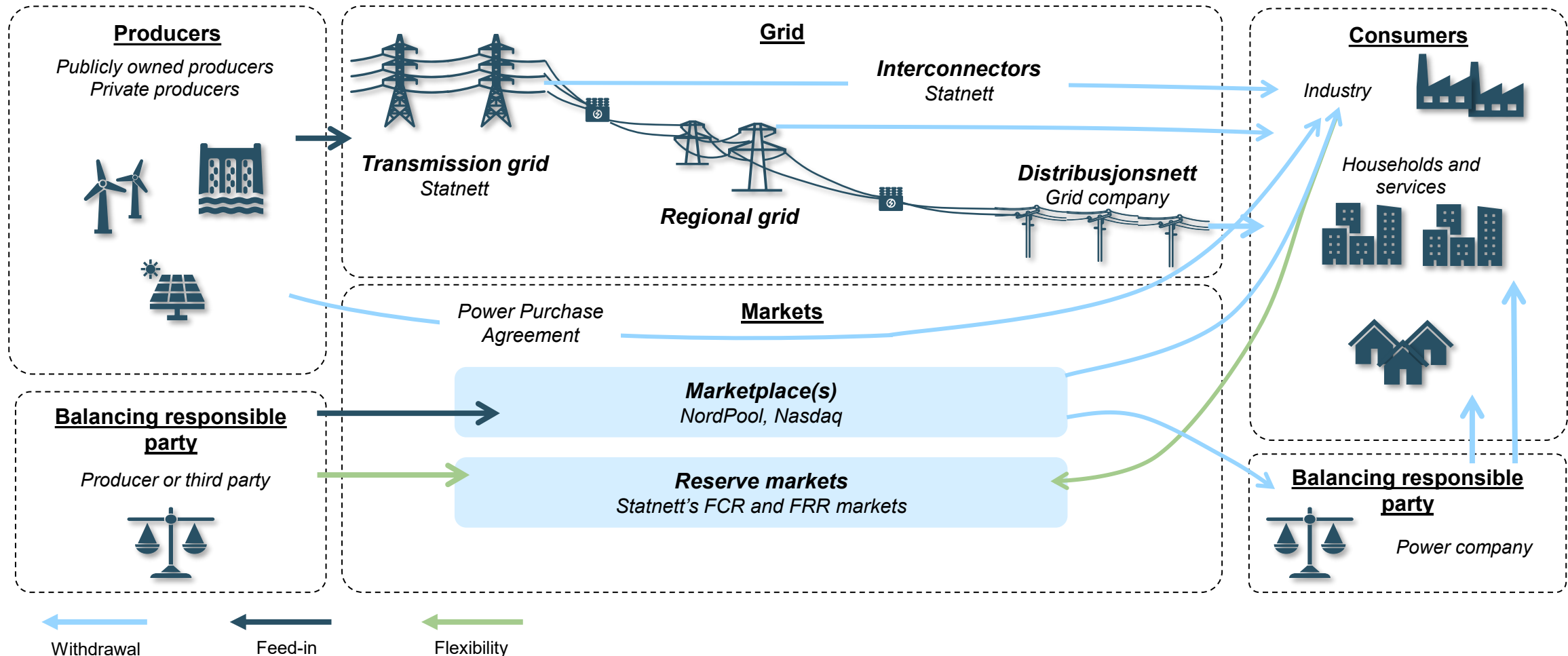
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The power system consists of producers, consumers and the grid – trade is cleared on a marketplace



Correct pricing is essential for an efficient power system, but achieving this requires a diverse array of instruments

Correct prices maximize social surplus

Supply

- Prices incentivize when and where to produce in the short term
- Prices signal where to investments in new generation capacity

Demand

- Prices incentivize when and where to buy electricity
- Prices signal where new demand is most profitable

Grid

- Network pricing ensures financing of necessary grid infrastructure
- Network pricing maintain grid reliability and sustainability

Finding correct prices is difficult (impossible?), several markets and instruments attempt to address this issue

- **Spot market** is where the power price is determined for most actors. One shared market clearing
- **Grid tariffs** ensure the grid is paid for and may provide price signals
- **Real-time markets** (*intraday markets, flexibility markets*) happen after the day-ahead market and are meant to fix differences between day-ahead volumes and actual dispatch
- **Capacity markets** ensure reliability and security of supply. They price in the value of available capacity [MW]
- **Financial instruments** and **bilateral contracts** price in risk and uncertainty, as well as expected future prices (*PPA's, futures, options, other derivatives*)
- **Environmental credits** and **subsidies** (seek to) price in the cost of emissions or value of renewables. (*RE certificates, EU ETS, CFDs and price premiums*)

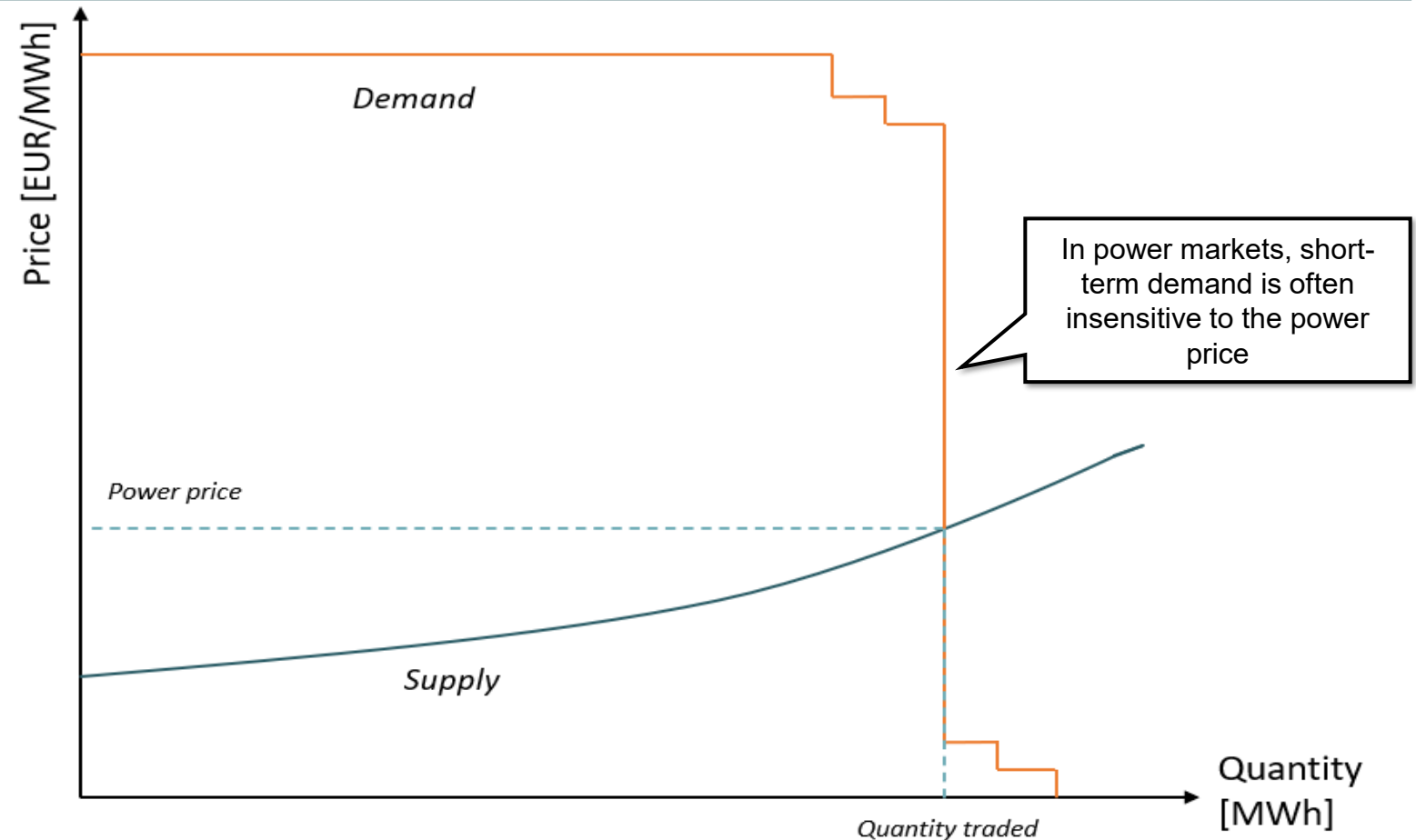
Economic theory:

Market-based trade yields the maximum social surplus (producer surplus and consumer surplus)

The price mechanism

- Social surplus is maximized in the **equilibrium** where the price equals the marginal production costs
 - The cheapest generation is chosen
 - Consumers pay less or equal to their willingness to pay
- All stakeholders seek to maximize private value
- One market clearing finds the optimal short-term price
- Also provides long-term price signals
- ***In practice several factors complicate the picture****

Example of a simple power market clearing



* Assumes perfect competition, perfect foresight and no grid costs, losses or bottlenecks

In practice:

Several factors cause inefficiencies, and various price mechanisms are added or adapted to handle them



Addresses problem



Adapted to problem

Price signal

Spot market

Real-time markets

Grid tariffs

Capacity markets

Financial instruments

Environmental credits and subsidies

	Grid infrastructure	Uncertainty	Market Power	Environmental Cost
	- Bottlenecks can cut off cheap production from demand - Losses affect optimal system price and allocation of production - The grid is itself expensive to build and operate	- Generation profiles and demand may change between market clearing and dispatch - Long-term and future prices are uncertain	- Participants with market power can affect prices and reduce market efficiency - Market power can be exacerbated by grid infrastructure	Market prices need to reflect environmental cost of the products (emissions, land use etc.)
Spot market	!		!	
Real-time markets		✓	!	
Grid tariffs	✓			
Capacity markets		✓		
Financial instruments		✓		✓
Environmental credits and subsidies				✓

Example:

Grid infrastructure adds complexity, requiring additional measures to prevent market inefficiencies

The power grid introduces two critical issues

1. It affects the optimal allocation of resources

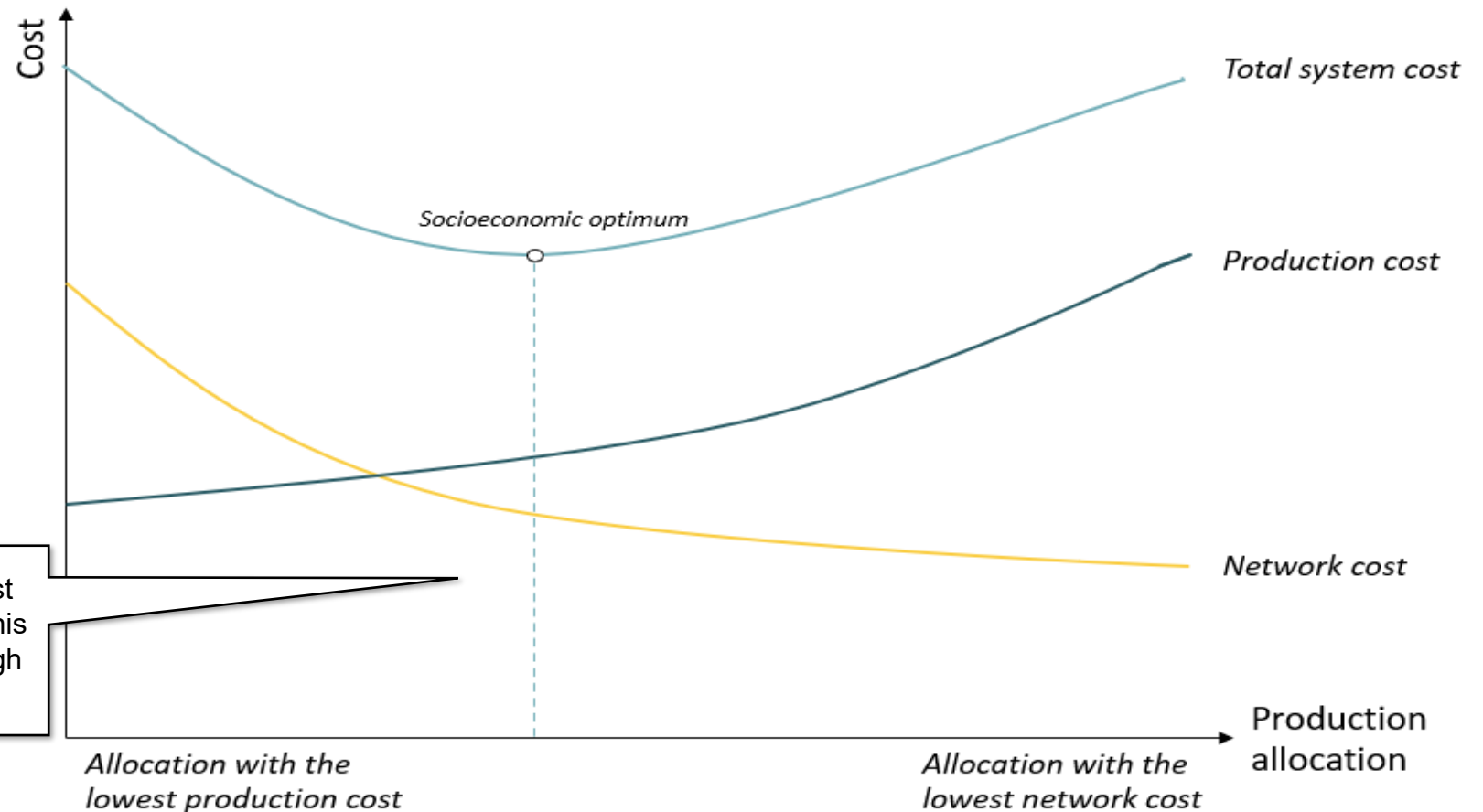
- Bottlenecks can cut off cheap production from demand
- Losses increase system costs and thus affect optimal system price
- Losses differ between locations and can change the optimal production allocation

2. It needs financing

- The grid is itself expensive to build and operate

Producers need to “see” or account for the grid cost they cause, for the system to approach optimum. This is done through price signals, communicated through **grid tariffs** and **differentiated market prices**.

Network costs affect optimal allocation of production



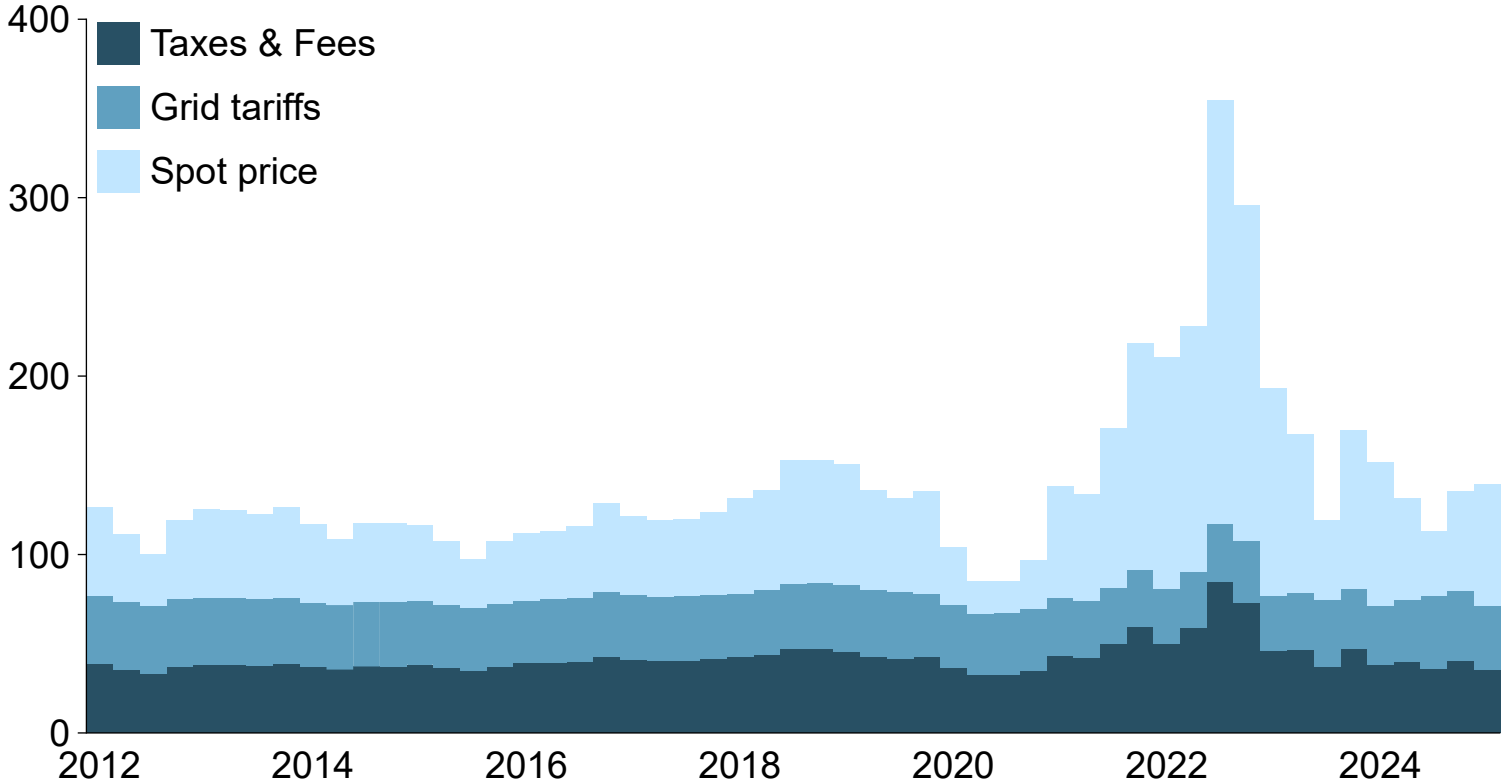
This presentation will cover the spot market and grid tariffs



They are the most important factors in the end-user electricity bill, which also consists of taxes and fees

The end-user price divided by components in current prices

End-user electricity costs per quarter (adjusted for inflation to Q1 2025)



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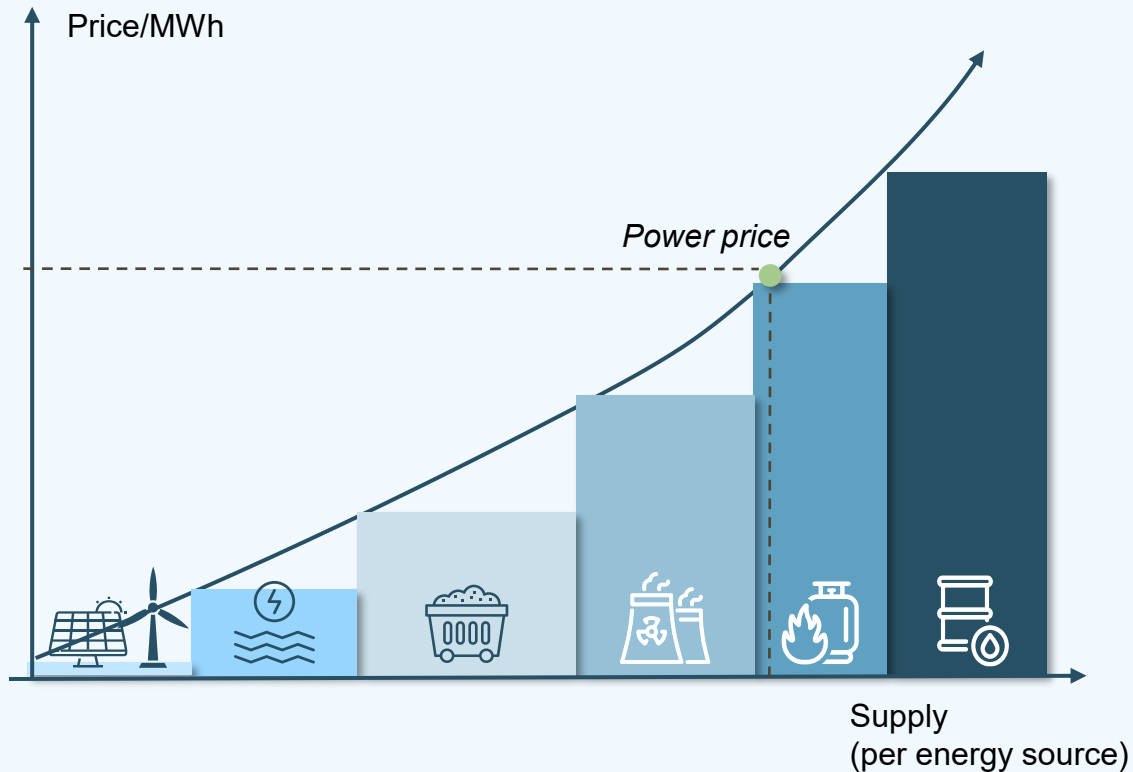
4 Grid tariffs



The spot market is the main market mechanism to manage production and consumption of electricity

Time-horizon	Manage risk	Manage energy		Manage system stability
	 10 years – 12 hours before delivery	 12-36 hours before delivery	 > 30 minutes before delivery	 at delivery
Market	Forward	Day-ahead (spot)	Intraday	Reserve/ balancing
Key properties	The financial market - Actors handle future price risk and speculate in future prices - Financial trade without physical settlement - Settlement services	Elspot - Market actors balance their physical positions - Operational planning and capacity allocation - Equilibrium between supply and demand is established for delivery each hour (quarter) the next day	Elbas/XBID - Intraday-markets with auctions and continuous trade until 30 minutes before delivery - Handles forecast errors for production and consumption and other deviations	Reguleringsmarkedet - The transmission system operator balances the system in real-time - Continuous physical balancing of supply and demand by the TSO to maintaining system frequency - Bottleneck management

Prices in the spot market are based on the **merit-order curve** where the **marginal cost sets the price** (The most expensive production source needed to cover the demand sets the price)



The spot market has a marginal pricing system

- » The merit order ranks suppliers/production technologies by their marginal costs/variable costs
- » The price is set by the marginal producer (the producer with the highest marginal/variable cost)
- » All producers receive the market clearing price and all consumers pay the market clearing price

The Nordic power system consists of many bidding zones, each with a different market price

The zonal configuration addresses grid bottlenecks

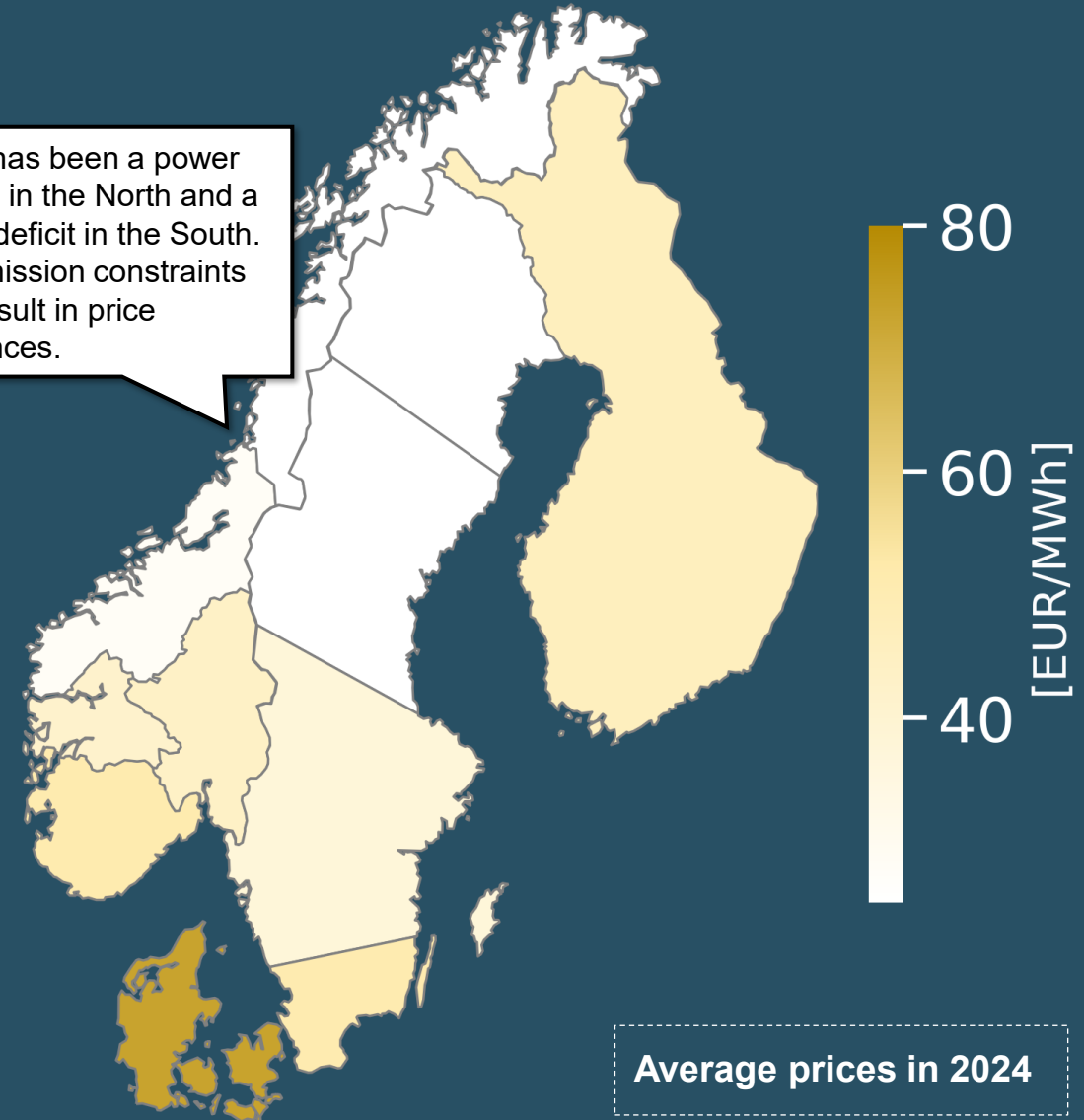
Bidding zones are used for congestion management

- Bidding zone borders should reflect locations where bottlenecks in the grid often occur
- Generators and consumers receive the price in the area in which their generation facility is located
- The TSO defines the bidding zone configuration. The Norwegian TSO (Statnett) has changed the delimitation multiple times, due to changing system needs

Bidding zones are a measure to obtain both short- and long-term efficiency

- In the short term, all market participants face correct alternative prices: Activating all adjustable resources to achieve an efficient dispatch. Market flows go from cheap to expensive zones*
- In the long-term, bidding zones incentivize efficient location of generation and consumption

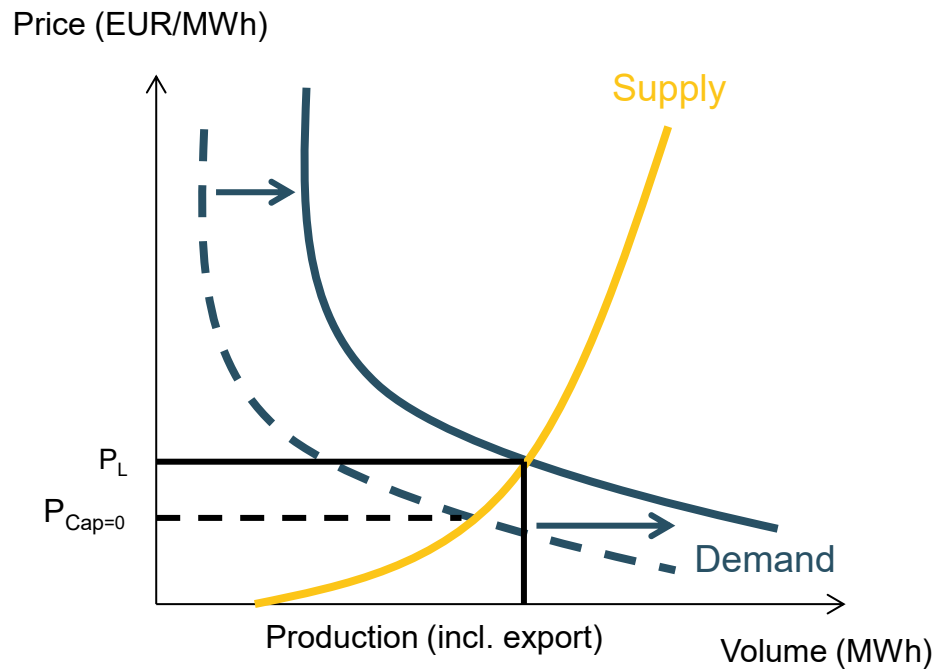
There has been a power surplus in the North and a power deficit in the South. Transmission constraints then result in price differences.



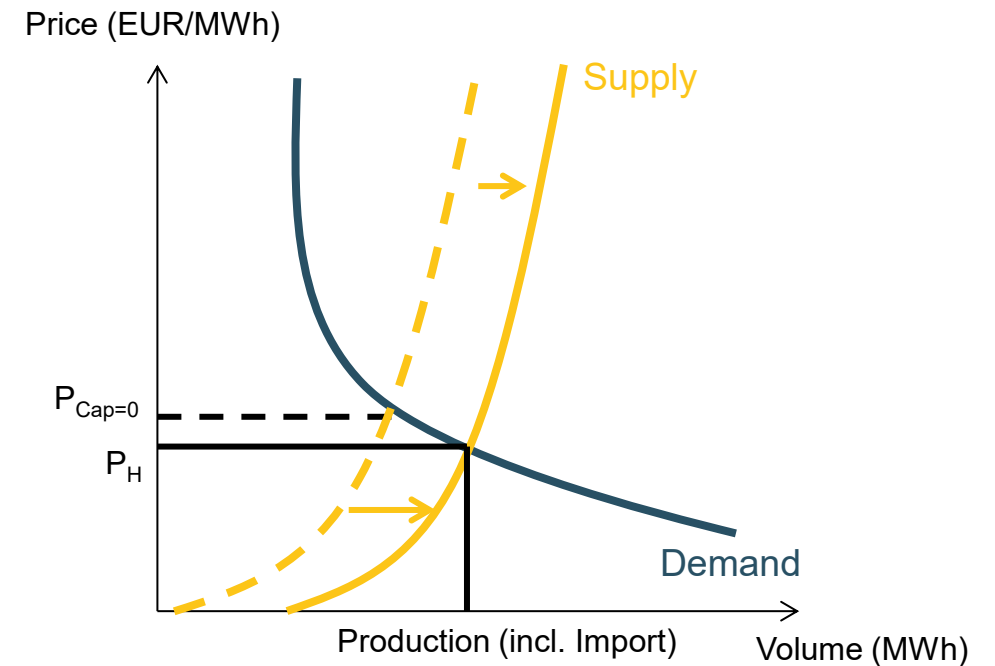
* The new flow-based market algorithm can allow counter-intuitive flows from a higher-price to a lower-price zone, but only if this improves the objective function (which maximize social surplus)

The area price accounts for grid constraints and is impacted by the net flow to/from other bidding zones

Surplus area

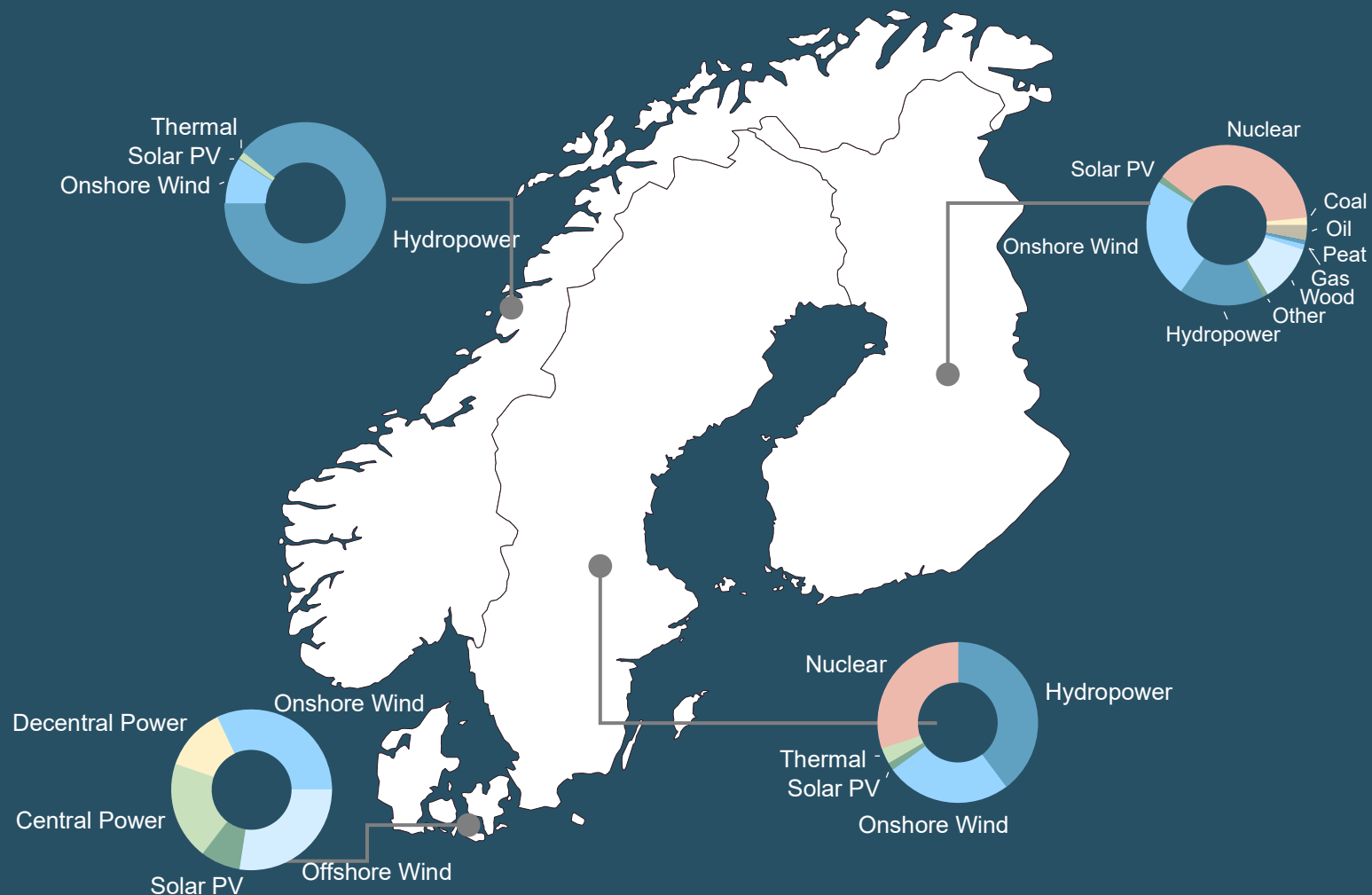


Deficit area



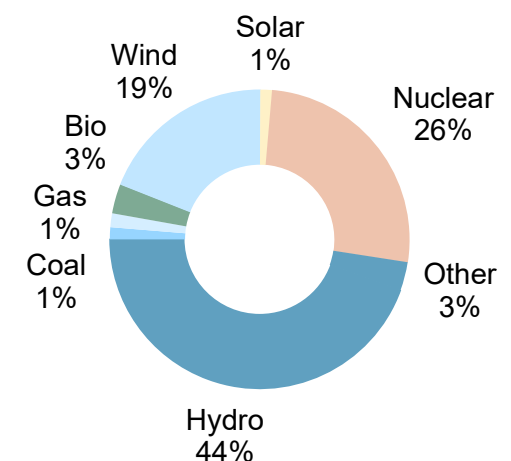
- The area price is found in the equilibrium point found after the inclusion of flows between areas
- The price in a surplus area is increased due to export and the price in the deficit area is decreased due to import
- Price coupling (same price between zones) occurs when the transmission capacity is non-binding

The Nordic power system consists of a large amount of weather-dependent production



Hydropower, wind and nuclear play large roles

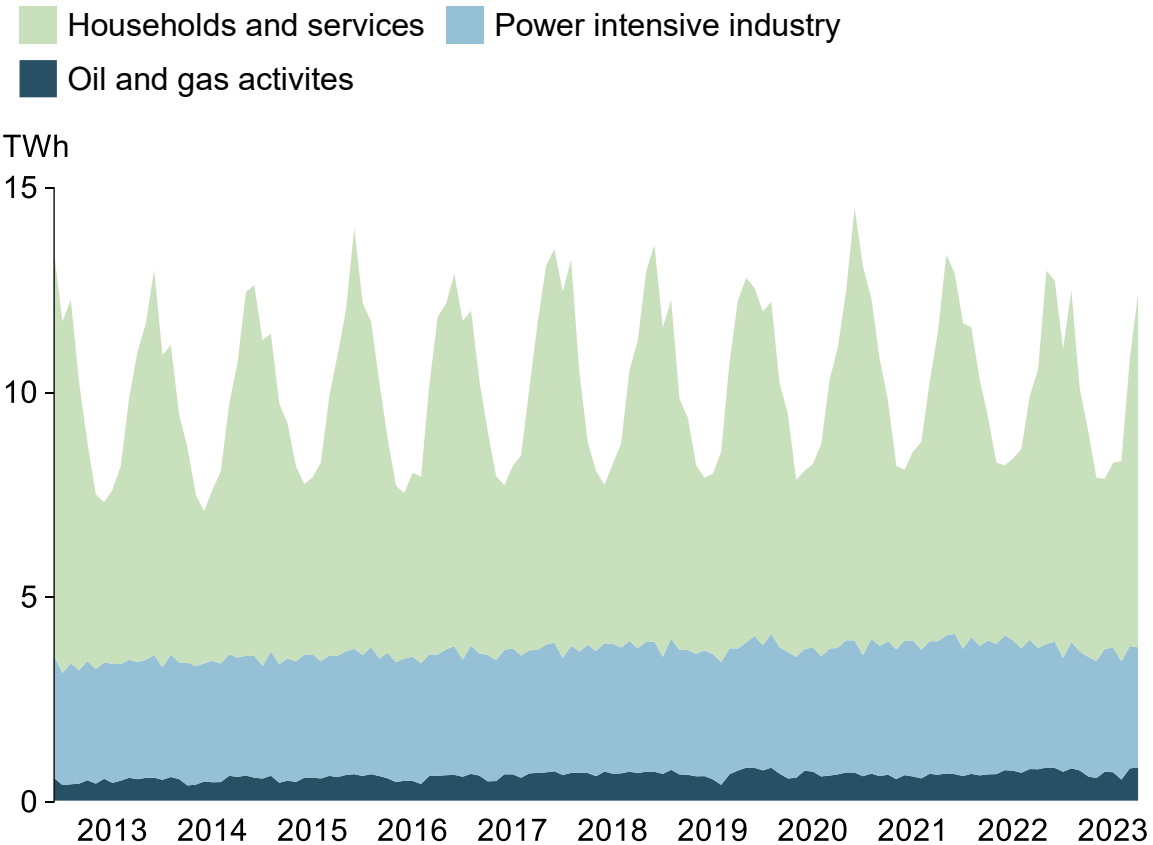
Total production mix for the Nordics, percentage



Source: Statnett (2025), Energinet (2025), Fingrid (2025), Svenska Kraftnät (2025), IEA (2024), Montel (2025)

A large portion of the power consumption stems from households and services...

Monthly electricity consumption in Norway (TWh/month)



.... yielding large daily and seasonal variations in both production and consumption

There is weather dependency in both production and consumption



Both wind power and hydropower is weather dependent – the production capacity depends on wind conditions and water inflow



Water reservoirs, especially prevalent in Norway, reduce some unpredictability. Still, much hydropower relies on small or nonexistent reservoirs, and rainfall can vary significantly between years



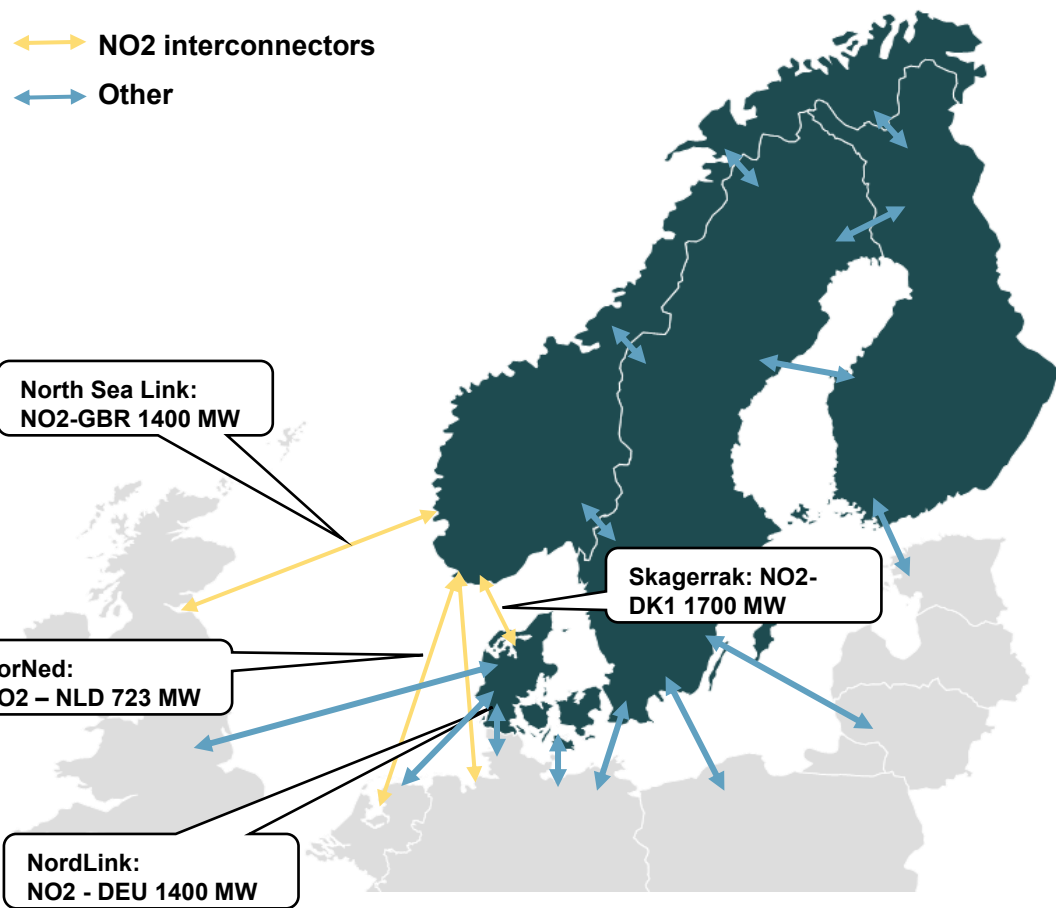
While hydro inflows are largest during the spring and summer, the demand is highest during winter. The electricity demand in Norway has strong seasonal variations, because of a large share of electrical heating.

The Norwegian power market is well integrated with the rest of the Nordics and Continental Europe

Interconnectors integrate the Nordic and the Continental power markets

↔ NO2 interconnectors

↔ Other



More power trade is beneficial for the total social surplus, but prices become influenced by other markets

Interconnectors help limit risk and increase the economic efficiency



Weather risk is limited when production is located across larger distances and has available transfer capacity inbetween.

Overcapacity and redundancy can be reduced in each zone, limiting costs.

With more interconnectors and less bottlenecks, we get more economically rational allocation of production.

Water values in hydropower reservoirs are crucial for the price formation

Water values are the expected value of water in the hydropower reservoirs, i.e., the opportunity cost of generation. It depends on the expected future prices, inflows to the reservoir, and the degree of flexibility in the hydropower plant.



The water values are largely given by prices in neighboring markets through the interconnectors, since the *opportunity cost* is influenced by the value of exporting to Continental Europe.

The actual short-run marginal costs of hydro generation are close to zero, yet the prices in the Nordic area are far from zero.

Water values are crucial for the Norwegian power price



The water value is the value/price hydropower producers bid into the spot market



Role: Ensure «optimal» usage of water resources throughout and between years



Interpretation: Opportunity cost of production – what is the cost of the alternative for providing the unit of electricity?



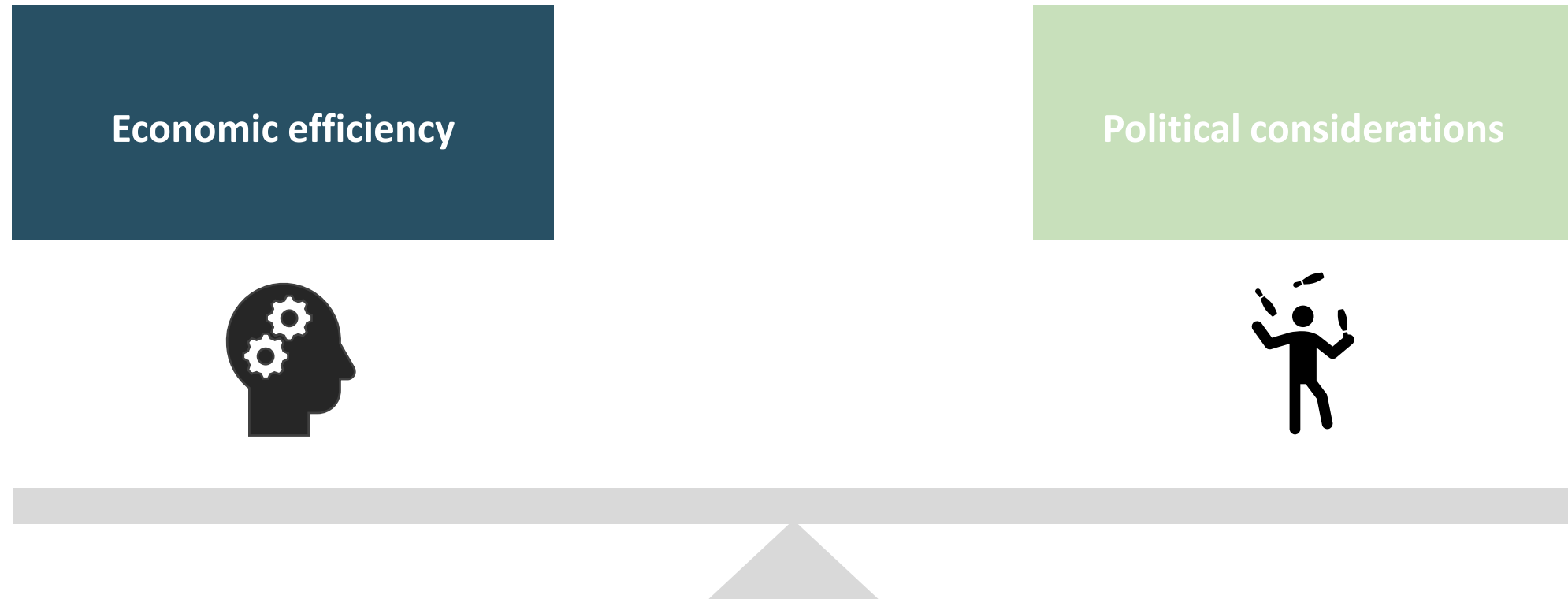
Challenge: The actual short-run marginal costs of hydropower production are very low



What would happen if hydropower producers bid into spot with a price close to zero?



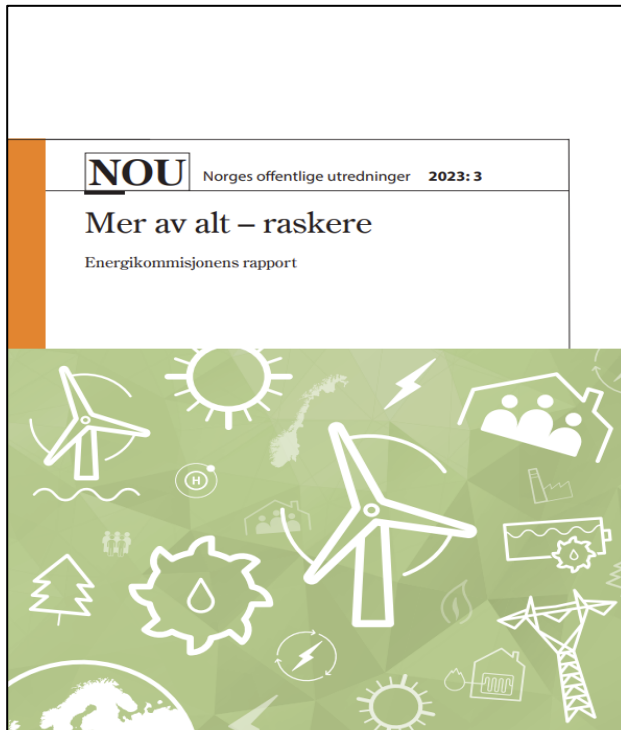
Political considerations influence policy and thus the power market



Example:

There are regularly conflicts between the power sector and environmental concerns

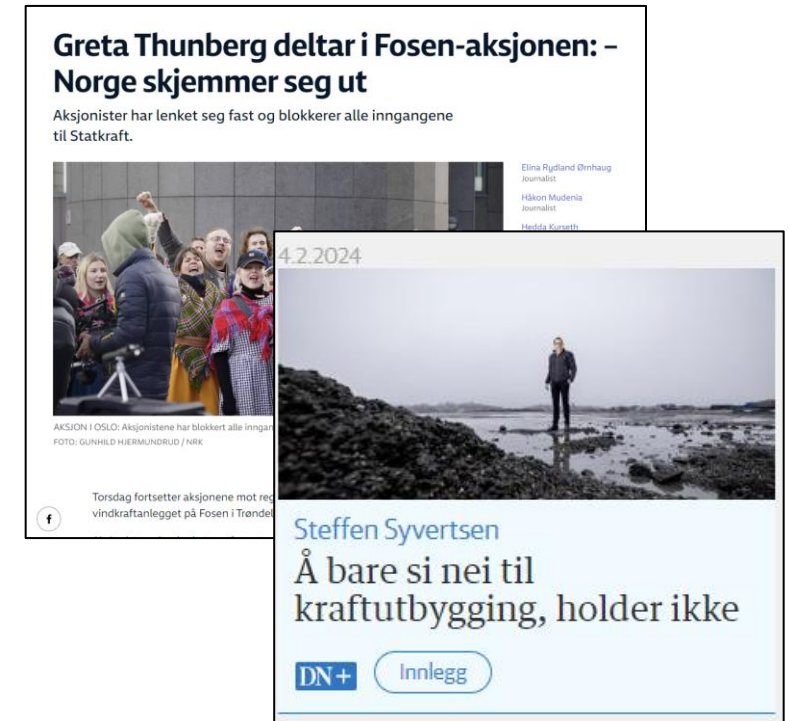
We need more power and a better grid



... but new build-out entails encroaching upon nature



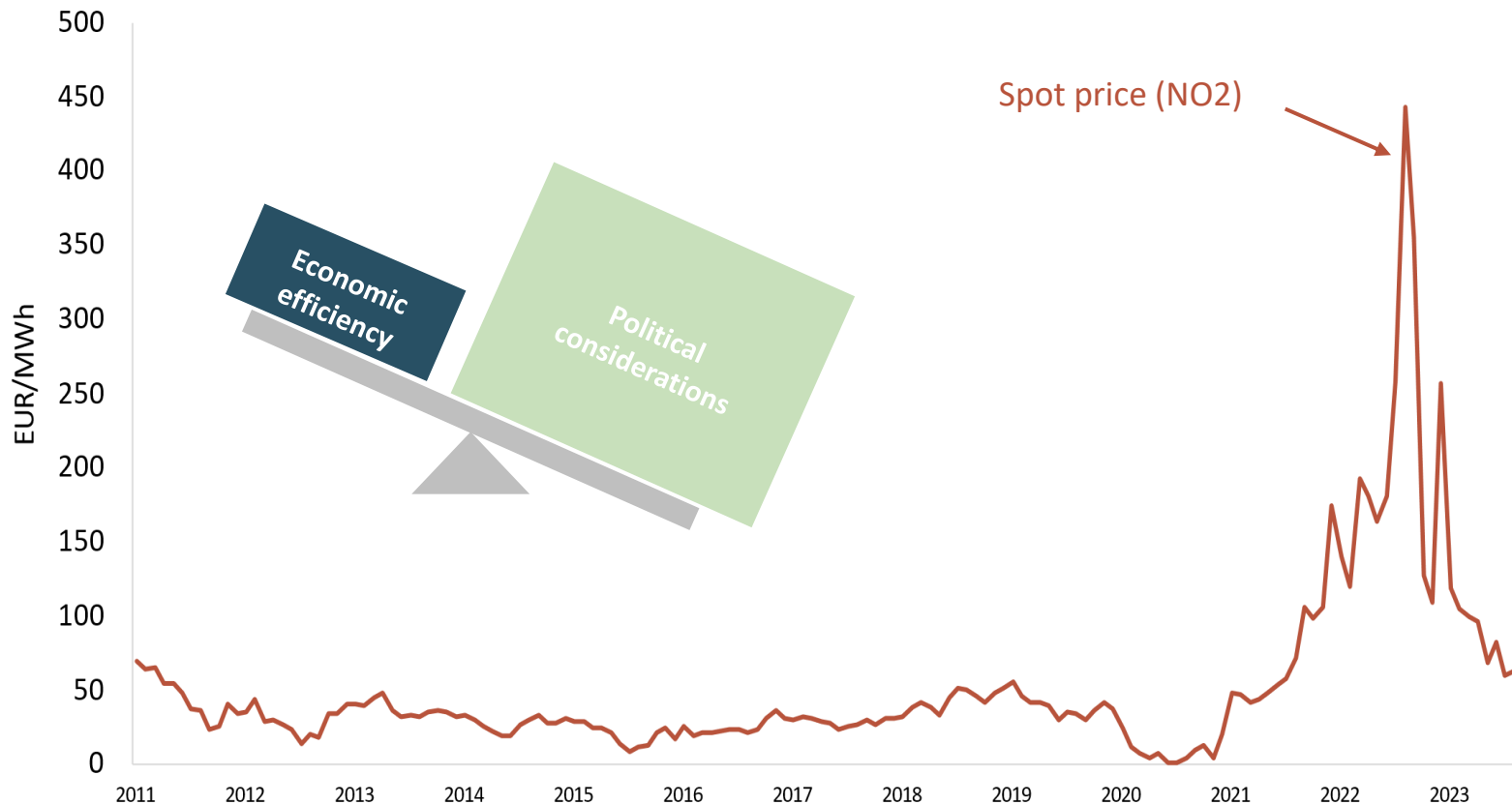
... leading to regular conflicts



Nature constitutes an externality, where the loss of nature is a cost not reflected in price signals. Consequently, political opposition to new developments, despite their apparent profitability, might be more beneficial for the overall system, including from an economic perspective.

Example: The 2022 price shock put power market pricing into question

Norwegian power prices soared in 2022

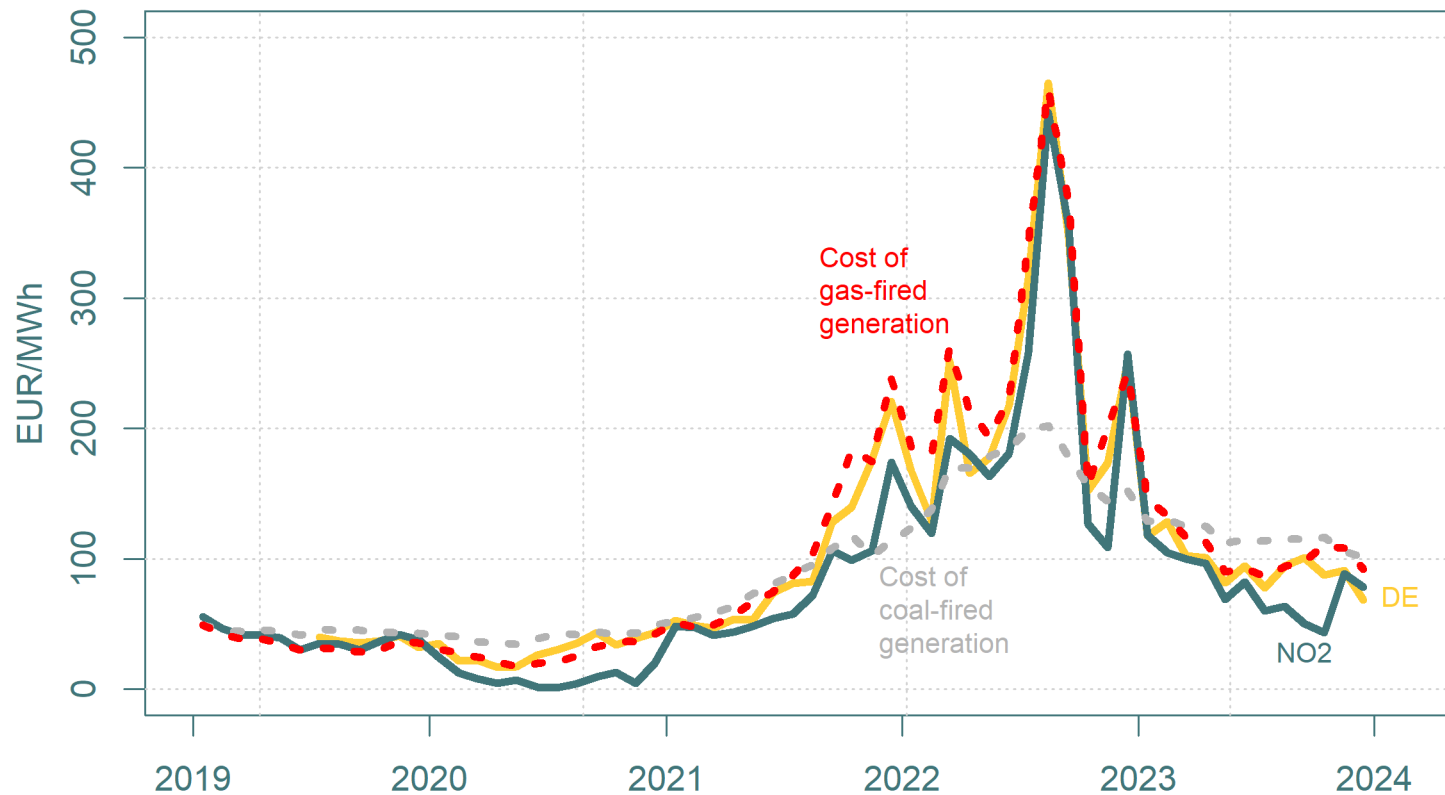


Source: Balansekunst - Strømprisutvalget

...and suddenly electricity pricing was a heated political issue

- **Why are prices high when costs are low?**
 - Marginal cost pricing was criticized
- **Are the large profits of power producers acceptable?**
 - Implemented extra taxes for producers
- **Should consumers pay so much for electricity?**
 - Implemented electricity subsidies for consumers
- **Why should Norway import high prices from the continent?**
 - Souring mood regarding interconnectors

The price spike in 2022 – 2023 was linked to key events



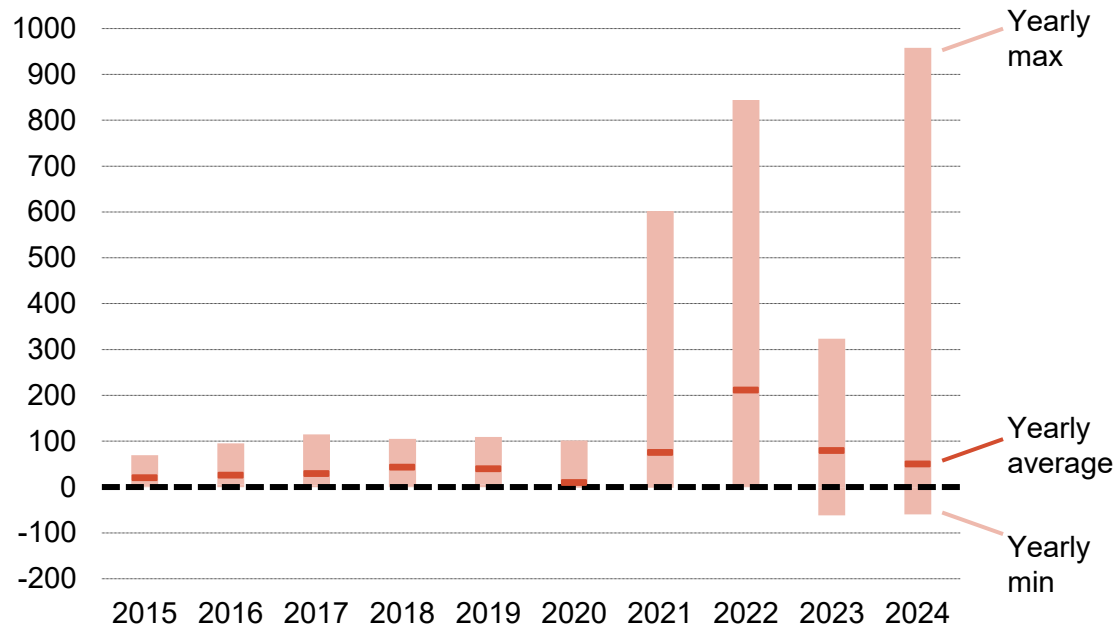
... higher gas prices and less supply

- 1 Russia choked the European gas supply leading up to and during the Ukraine war, leading to significant increases in gas prices.
- 2 Norwegian hydropower experienced a dry year
- 3 France, a major power exporter in normal years, also had to shut down nuclear power due to maintenance and drought

*The Short-Run Marginal Cost of a gas-fired power plant with 50% efficiency.

Now: Volatility has skyrocketed in Southern Norway

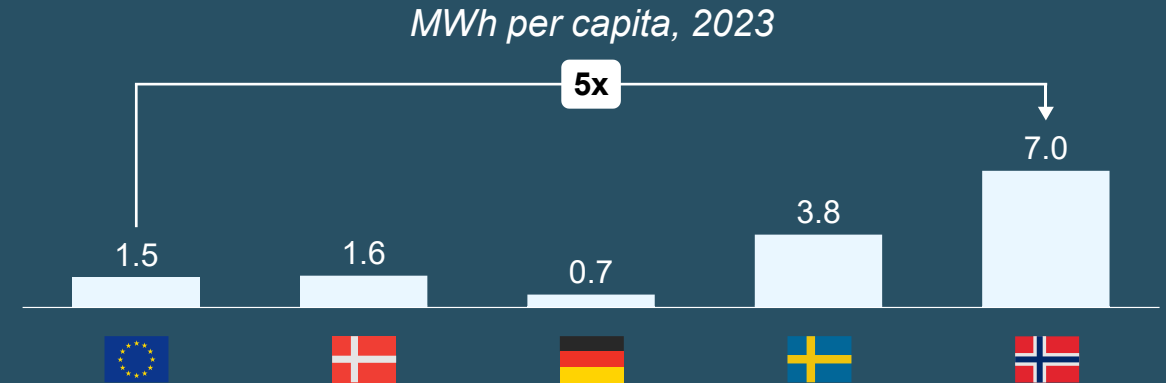
Yearly prices in NO2, nominal EUR/MWh



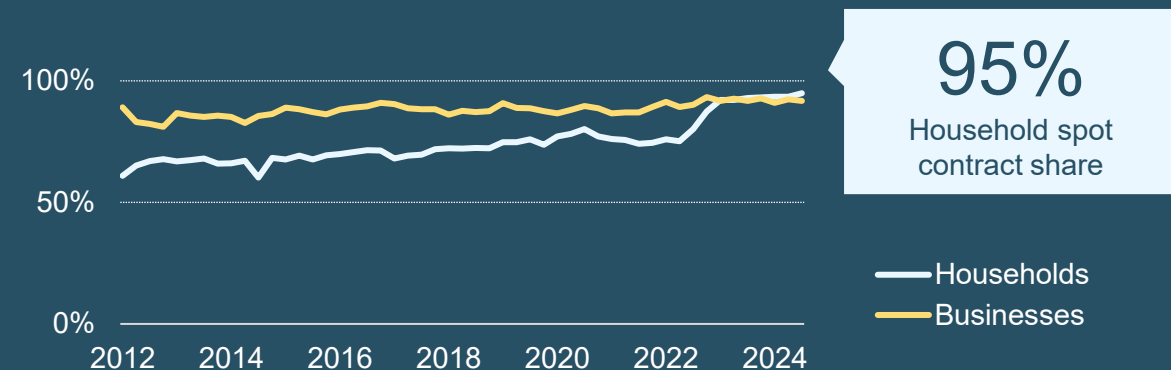
- › Historically Norway has had low and stable power prices due to a power surplus and significant amounts of flexible hydropower (50% of Europe's flexible hydropower)
- › The power surplus has enabled competitive energy-intensive industry and electricity-based household heating.

... while the Norwegian electorate is highly exposed

Household power use 5x higher in Norway than in the EU



Most households and businesses pay hourly wholesale prices



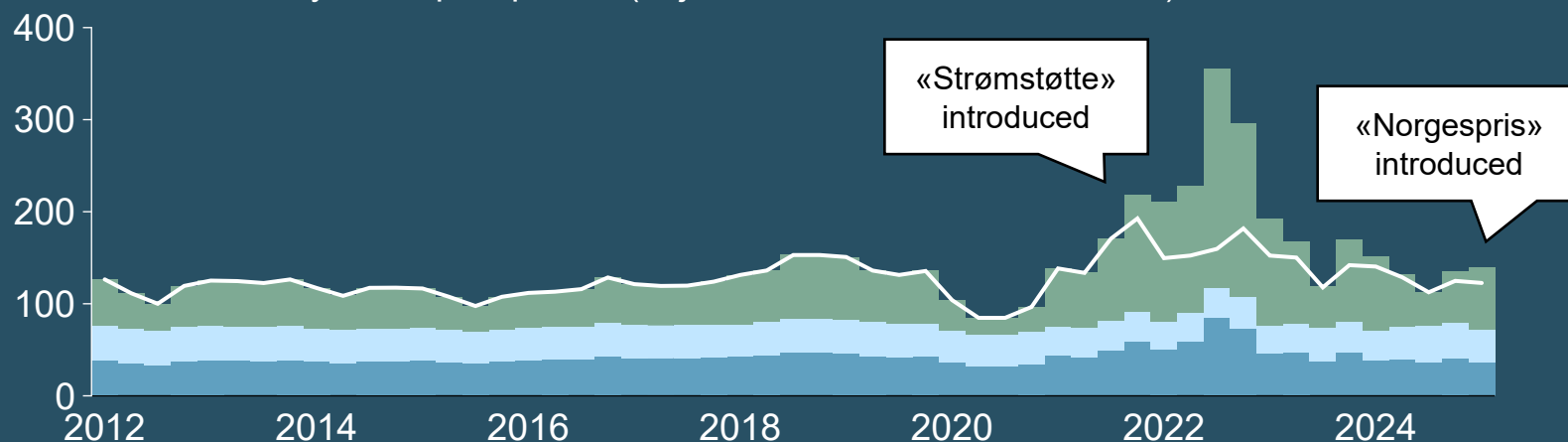
Source: ENTSO-e (2025), SSB (2025), Eurostat (2024)

The «Package for Lower Electricity Prices and Control Over Power» was introduced as a response

The package offers a new subsidy scheme for household consumers

- 1 Fixed power prices of 40 øre/kWh (excl. VAT) for households (Norgespris)
- 2 Reduced VAT (mva.) on grid fees, from 25% to 15%
- 3 The government asks the TSO not to plan any new subsea interconnectors until the end of the next parliamentary term in 2029.

End-user electricity costs per quarter (adjusted for inflation to Q1 2025)



Støre's electricity plan: Fixed price agreements of 40 øre/kWh

The Labour Party presents its new current policy on the first day after its divorce from the Centre Party.



At 11 a.m., Prime Minister Jonas Gahr Støre and Minister of Energy Terje Aasland presented new electricity measures.

PHOTO: OLE BERG-RUSTEN / NTB

[Ragnhild Vartdal](#)

Journalist

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4 **Grid tariffs**



The TSO Statnett and multiple local DSOs are responsible for building grid capacity in Norway



Regulator

Transmission grid
132-420 kV*

Statnett is the Transmission System Operator

- Operates the transmission grid
- Owns >95 % of the transmission grid, in the process of taking over the remainder

Regional grid
33-132 kV

56 network owners (incl. power companies and industry)

- Most own distribution grids as well
- Licence for each individual network asset

Distribution grid
<22 kV

77 network owners (incl. power companies and industry)

- Differ in size from >900 000 customers to a handful
- Area licences with local grid monopoly

*Source: RME. *All assets >200 kV is classified as central transmission grid. In addition, RME will process individual applications from regional grid owners to reclassify network facilities. This applies to 132 kV lines that will be included in the transmission grid if they are of significant importance for the operation of the transmission system. The number of companies at each level is based on RME's revenue cap notification for 2023. THEMA Consulting Group*

The grid is a natural monopoly that needs to be regulated to avoid inefficiency and suboptimal use of the grid



The grid is a natural monopoly

The electricity grid is a natural monopoly in the sense that it is **economically inefficient to have parallel grids in an area**

The grid must be regulated to ensure rational operation and development

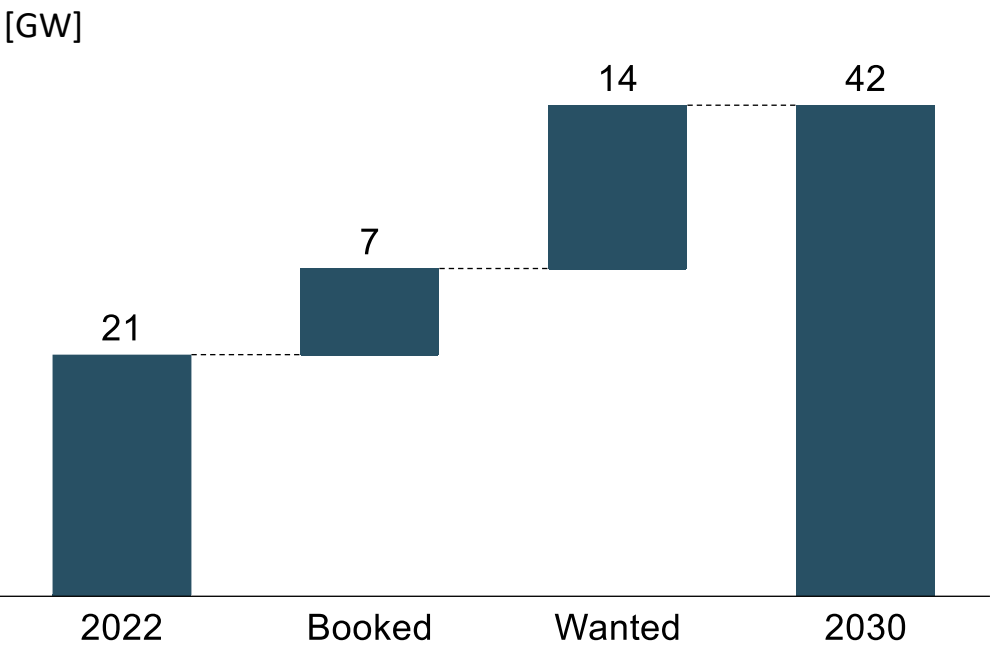
Regulation is intended to **prevent the network companies from exploiting their monopoly position** and charging too much, and ensure the best possible power grid and supply at the lowest possible price

The grid companies' income is paid by the customers through tariffs

The **income regulation framework determines the income** each company can collect. The allowed income is **distributed among all users of the network** and collected through the grid tariffs

There is an increasing demand for grid capacity, which requires large investments in the grid, and the investments must be financed

Delivered and potential new demand for power capacity in the Norwegian grid (regional+distributional)



Based on a project for the 12 largest grid companies in Norway



A rough estimate of 150 billion NOK in investments to cover the needs until 2030

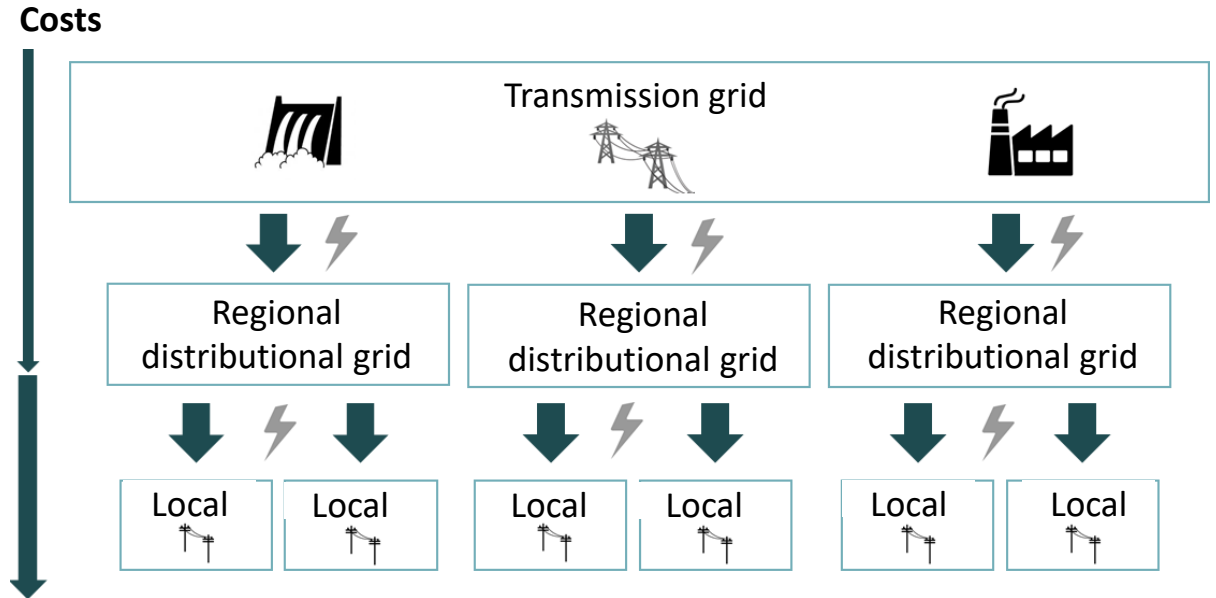


The (Norwegian) tariff model

Consists of the following elements

Energy charge	• Charge per kWh of generation or consumption per node • Cover costs of energy losses in the grid
Fixed charge	• Differentiated according to power (MW) • Covers customer-specific costs in addition to a share of other fixed costs related to grid operation
Connection charges	• One-off payment for customer-specific grid investments, both new assets and reinforcement of existing connections

... and sums up the cost from all grid levels the customer uses



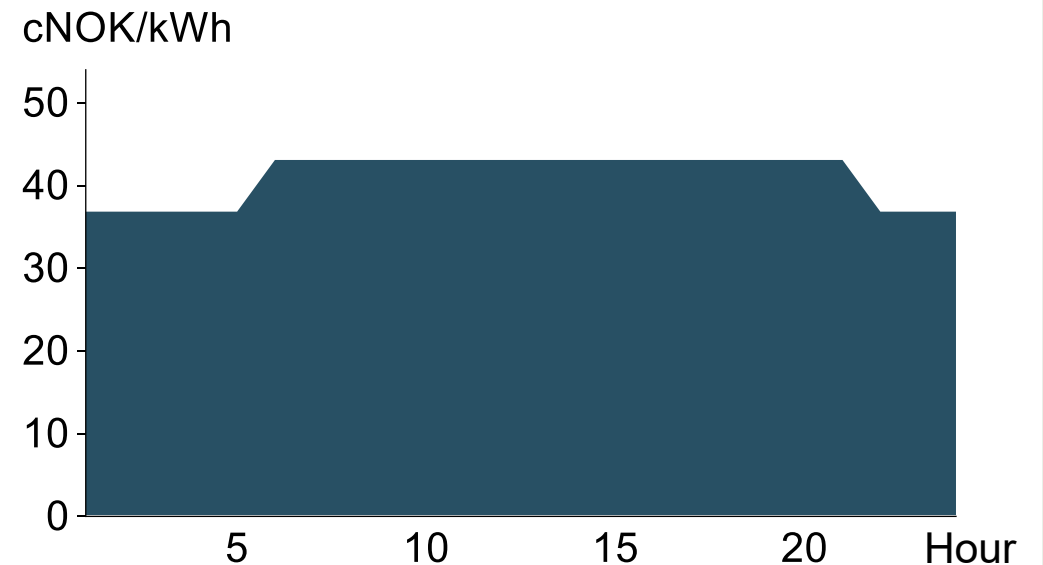
The energy part incentivize lower energy consumption, and moving the usage to the night hours (time of use)

Energy based tariff element

- Cost per kWh (energy)
- Covers marginal losses in the grid
- Can cover other fixed costs
- Can at most make up 50% of DSO's income
- Can be differentiated by time of use

Incentive: Use less energy

The cost for energy has a «time of use» differentiation with lower costs at night and during weekends



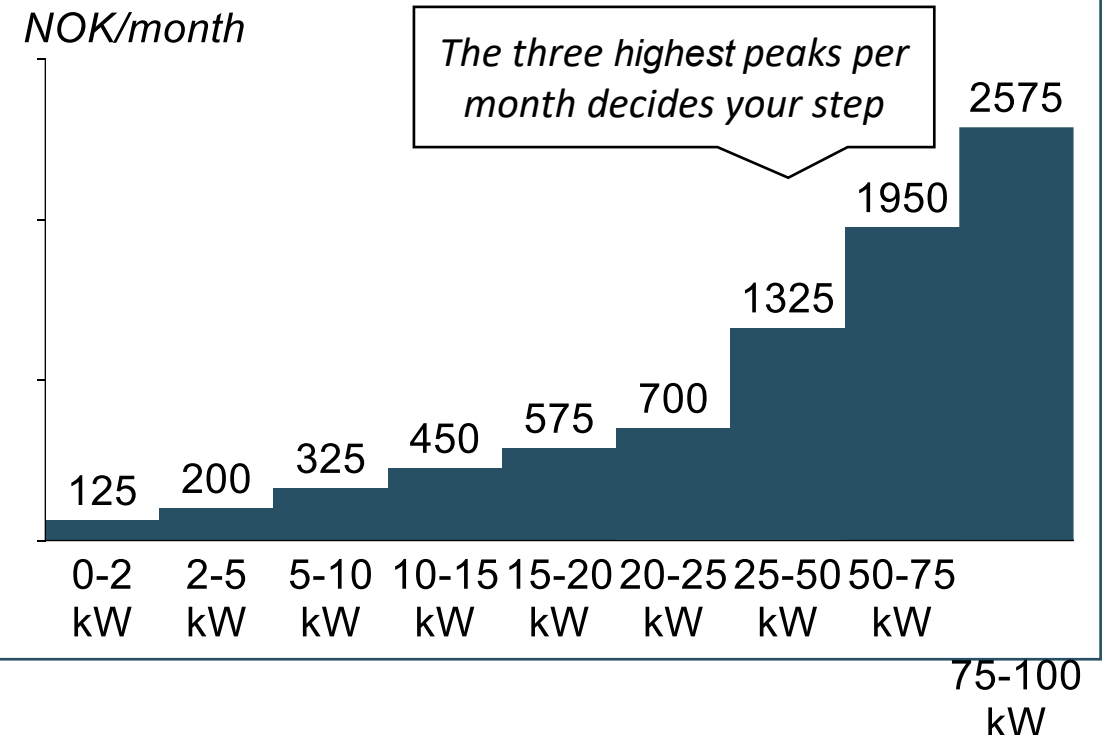
... while the power part incentivize lower peaks through the “power staircase”

Power based tariff element

- Cost per month
- Must be differentiated based on power demand
- Must cover customer specific costs (Billing, measuring..)
- Can cover other fixed costs

Incentive: Use less energy at the same time

The power cost per month is decided through placement in the «power staircase»



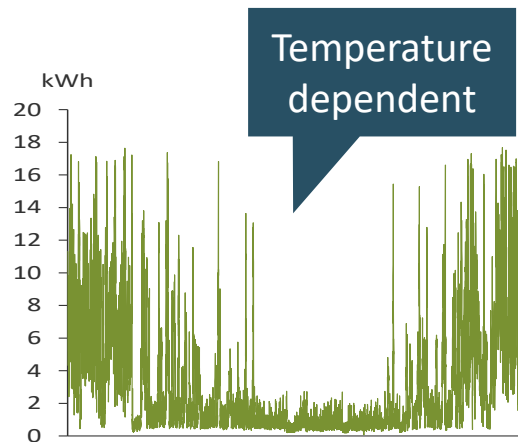
Example: we looked at three cases to see how the power tariff affects them:
a baseline with no EV-charging, a fast EV-charging and a slow EV-charging

Baseline (no car)



Consumption: 27 000 kWh

Annual consumption profile



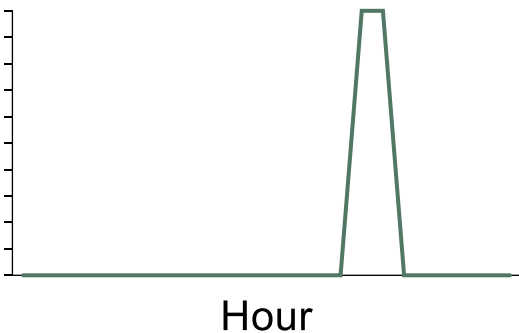
Milage: 0.2 kWh/km

Fast charging



Charging: 2400 kWh = 12 000 km

**Daily charging profile
(1 hour at 17:00)**

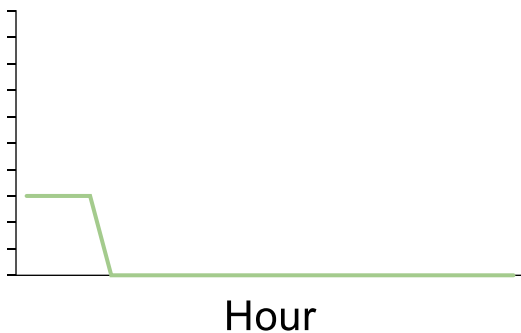


Slow charging



Charging: 2400 kWh = 12 000 km

**Daily charging profile
(3 hours from 00:00 to 03:00)**



BASELINE: Our house is heavily affected by the outdoor temperature, with high power peaks in the winter

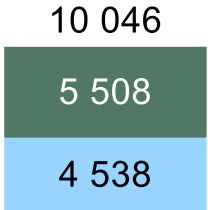
Energy consumption (kWh/year)



Baseline (No car)
27 000 kWh

Yearly tariff (NOK/year)

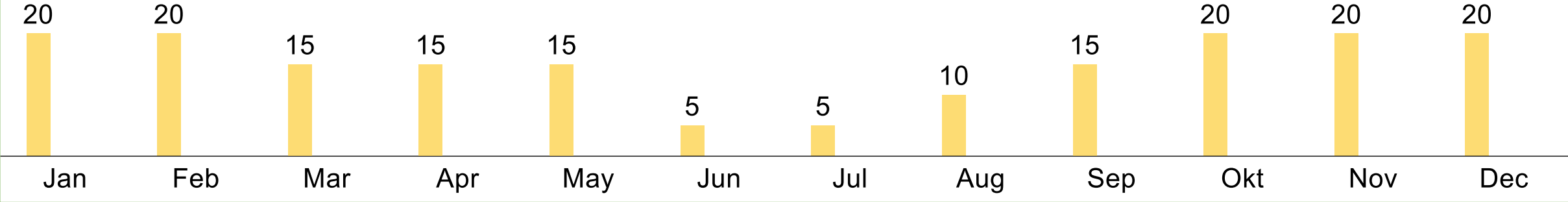
Energy Power



We're only looking at the grid-part of the bill, the electricity bill will come in addition.

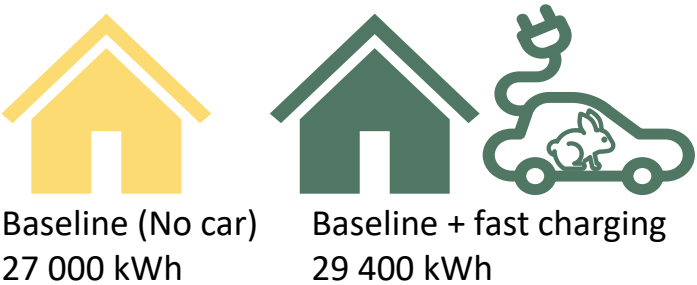
Baseline (no car)

How much capacity is paid for? (kW)
(Average of three highest peaks per month)

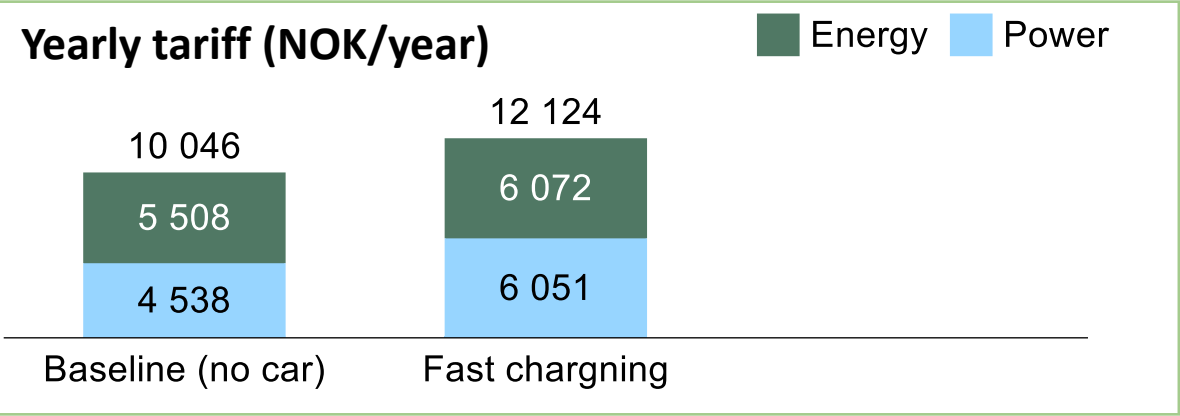


CASE 1: We buy an EV and charge it for 1 hour every afternoon. Every month our power consumption jumps up to the next step in the «power staircase» compared to the baseline

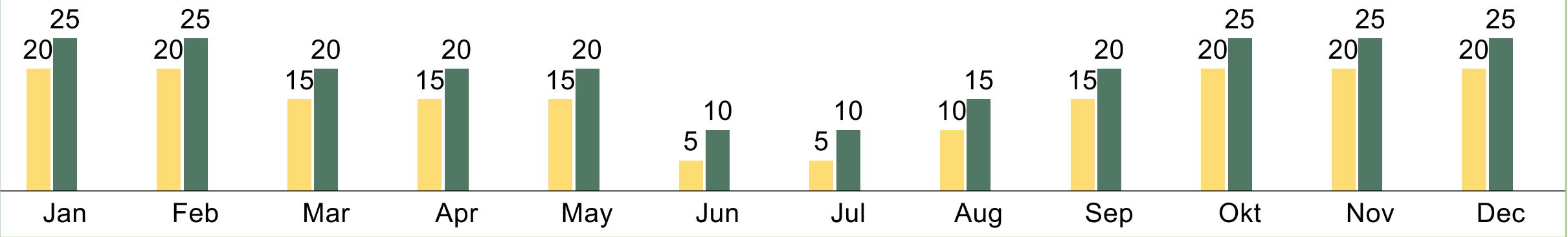
Energy consumption (kWh/year)



Yearly tariff (NOK/year)

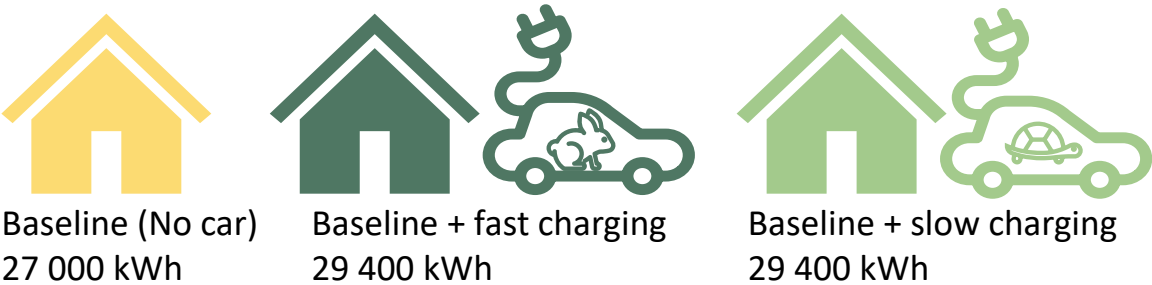


How much capacity is paid for? (kW)
(Average of three highest peaks per month)

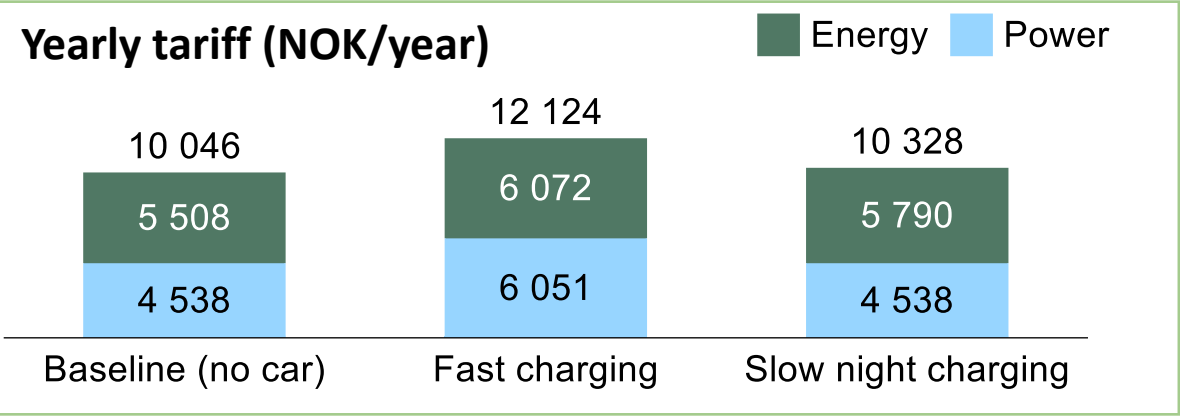


CASE 2: We move our EV-charging to the night hours and slow it down. Compared to the baseline, we only have to pay grid rent for more energy, not more power to charge our EV.

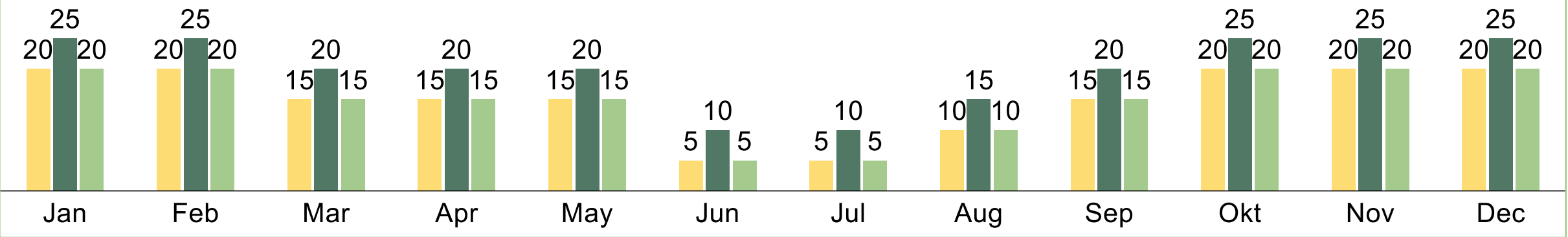
Energy consumption (kWh/year)



Yearly tariff (NOK/year)



How much capacity is paid for? (kW)
(Average of three highest peaks per month)



Summary



Spot market pricing

- Pricing in the spot market is based on the merit-order curve. This yields marginal pricing and aims at maximizing the social surplus (producer and consumer surplus).
- Pricing is complicated by several factors: grid infrastructure, uncertainty, market power and environmental cost



Grid infrastructure and tariffs

- Grid tariffs are designed to reflect the cost of building and maintaining the grid, costs that are not included in the spot price of electricity.
- Grid tariffs are used to finance the expansion of transmission capacity.
- Grid tariffs consist of both an energy-based component and a power-based component.



Other

- Political considerations and policy influences the power market and price signals



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confidence